

# OS LAB TASK 11: Page Replacement Algorithms

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Number of Pages = 12

Pages = 2 5 1 0 1 0 0 1 0 1 5 2

**First In First Out: ( Frames = 3 )**

Manual Implementation:

First In First Out (3 frames)

Pages: 2 5 1 0 1 0 0 1 0 1 5 2

|   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 2 | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 |   | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 2 | 2 |
| 3 |   |   | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
|   | f | f | f | f | h | h | h | h | h | h | f |

Page faults = 5  
Page hits = 7  
Total = 12

Hit Ratio =  $7/12 = 58.33\%$   
Fault/mis Ratio =  $5/12 = 41.67\%$

Code and Output:

```
bhaavya@bhaavya-VirtualBox: ~/Desktop/25MCB1010
#include <stdio.h>
#include <stdbool.h>
int main() {
    int n;
    printf("Enter the number of pages: ");
    scanf("%d", &n);
    int pages[n];
    printf("Enter the pages: ");
    for (int i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }
    int f;
    printf("Enter the number of frames: ");
    scanf("%d", &f);
    int frames[f];
    for (int i = 0; i < f; i++) {
        frames[i] = -1;
    }
    int faults = 0, hits = 0;
    int start = 0;
    for (int i = 0; i < n; i++) {
        bool flag = false;
        for (int j = 0; j < f; j++) {
            if (frames[j] == pages[i]) {
                hits++;
                flag = true;
                break;
            }
        }
        if (!flag) {
            frames[start] = pages[i];
            start = (start + 1) % f;
            faults++;
        }
        printf("%d: ", pages[i]);
        for (int j = 0; j < f; j++) {
            if (frames[j] != -1)
                printf("%d ", frames[j]);
            else
                printf("- ");
        }
        printf("\n");
    }
    printf("\nNumber of hits: %d\n", hits);
    printf("Number of faults: %d\n", faults);
}
```

```
bhaavya@bhaavya-VirtualBox:~/Desktop/25MCB1010$ vim fifo.c
bhaavya@bhaavya-VirtualBox:~/Desktop/25MCB1010$ gcc fifo.c -o fifo.exe
bhaavya@bhaavya-VirtualBox:~/Desktop/25MCB1010$ ./fifo.exe
Enter the number of pages: 12
Enter the pages: 2 5 1 0 1 0 0 1 0 1 5 2
Enter the number of frames: 3
2: 2 - -
5: 2 5 -
1: 2 5 1
0: 0 5 1
1: 0 5 1
0: 0 5 1
0: 0 5 1
1: 0 5 1
0: 0 5 1
1: 0 5 1
5: 0 5 1
2: 0 2 1

Number of hits: 7
Number of faults: 5
bhaavya@bhaavya-VirtualBox:~/Desktop/25MCB1010$
```

**Least Recently Used: ( Frames = 3 )**

Manual Implementation:

Least Recently Used (3 frames)

Pages = 2 5 1 0 1 0 0 1 0 1 5 2

|   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 2 | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 2 |
| 2 |   | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| 3 |   |   | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
|   | f | f | f | f | h | h | h | h | h | h | f |

Page faults = 5  
Page hits = 7  
Total = 12

Hit Ratio =  $7/12 = 58.33\%$   
Fault Ratio =  $5/12 = 41.67\%$

Code and Output:

```
bhaavya@bhaavya-VirtualBox: ~/Desktop/25MCB1010
#include <stdio.h>
#include <stdbool.h>
int main() {
    int n;
    printf("Enter the number of pages: ");
    scanf("%d", &n);
    int pages[n];
    printf("Enter the page numbers: ");
    for (int i = 0; i < n; i++)
        scanf("%d", &pages[i]);
    int frame;
    printf("Enter the number of frames: ");
    scanf("%d", &frame);
    int frames[frame], recent[frame];
    for (int i = 0; i < frame; i++) {
        frames[i] = -1;
        recent[i] = -1;
    }
    int hit = 0, fault = 0;
    for (int i = 0; i < n; i++) {
        int page = pages[i];
        bool found = false;
        for (int j = 0; j < frame; j++) {
            if (frames[j] == page) {
                hit++;
                found = true;
                break;
            }
        }
        if (!found) {
            fault++;
            int replace_index = -1;
            for (int j = 0; j < frame; j++) {
                if (frames[j] == -1) {
                    replace_index = j;
                    break;
                }
            }
            if (replace_index == -1) {
                int lru_index = 0;
                int lru_pos = i;
                for (int j = 0; j < frame; j++) {
                    int k;
                    for (k = i - 1; k >= 0; k--) {
                        if (pages[k] == frames[j])
                            break;
                    }
                    if (k < lru_pos) {
                        lru_pos = k;
                        lru_index = j;
                    }
                }
                replace_index = lru_index;
            }
            frames[replace_index] = page;
            printf("%d: ", page);
        }
        for (int j = 0; j < frame; j++) {
            if (frames[j] == -1)
                printf("- ");
            else
                printf("%d ", frames[j]);
        }
        printf("\n");
    }
    printf("\nTotal Hits: %d\n", hit);
}
```

```
bhaavya@bhaavya-VirtualBox:~/Desktop/25MCB1010$ gcc lru.c -o lru.exe
bhaavya@bhaavya-VirtualBox:~/Desktop/25MCB1010$ ./lru.exe
Enter the number of pages: 12
Enter the page numbers: 2 5 1 0 1 0 0 1 0 1 5 2
Enter the number of frames: 3
2: 2 - -
5: 2 5 -
1: 2 5 1
0: 0 5 1
1: 0 5 1
0: 0 5 1
0: 0 5 1
1: 0 5 1
0: 0 5 1
5: 0 5 1
2: 2 5 1

Total Hits: 7
Total Faults: 5
```

**Most Recently Used: ( Frames = 4 )**

Manual Implementation:

Most Recently Used (4 frames)

Pages = 2 5 1 0 1 0 0 1 0 1 5 2

|   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 2 |   | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| 3 |   |   | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 4 |   |   |   | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|   | f | f | f | f | h | h | h | h | h | h | h |

Page faults = 4  
Page Hits = 8  
Total = 12

Hit Ratio =  $8/12 = 66.67\%$   
Fault Ratio =  $4/12 = 33.33\%$

Code and Output:

```
bhaavya@bhaavya-VirtualBox: ~/Desktop/25MCB1010
#include <stdio.h>
#include <stdbool.h>

int main(){
    int n;
    printf("Enter the number of pages: ");
    scanf("%d",&n);
    printf("Enter the page numbers: ");
    int pages[n];
    for(int i=0;i<n;i++)scanf("%d",&pages[i]);
    int frame;
    printf("Enter the number of frames: ");
    scanf("%d",&frame);
    int frames[frame];
    for(int i=0;i<frame;i++)frames[i]=-1;
    int hit=0,fault=0;
    for(int i=0;i<n;i++){
        int page=pages[i];
        bool found=false;
        for(int j=0;j<frame;j++){
            if(frames[j]==page){hit++;found=true;break;}
        }
        if(!found){
            fault++;
            int replace=-1;
            for(int j=0;j<frame;j++){
                if(frames[j]==-1){replace=j;break;}
            }
            if(replace==-1){
                int recent[frame];
                for(int j=0;j<frame;j++){
                    recent[j]=-1;
                    for(int t=i-1;t>=0;t--){
                        if(frames[j]==pages[t]){recent[j]=t;break;}
                    }
                }
                int max_index=0;
                for(int j=1;j<frame;j++){
                    if(recent[j]>recent[max_index])max_index=j;
                }
                replace=max_index;
            }
            frames[replace]=page;
        }
        printf("%d ",page);
        for(int j=0;j<frame;j++){
            if(frames[j]==-1)printf("- ");
            else printf("%d ",frames[j]);
        }
        printf("\n");
    }
    printf("Hits: %d\n",hit);
    printf("Faults: %d\n",fault);
    return 0;
}
```

```
bhaavya@bhaavya-VirtualBox:~/Desktop/25MCB1010$ vim mru.c
bhaavya@bhaavya-VirtualBox:~/Desktop/25MCB1010$ gcc mru.c -o mru.exe
bhaavya@bhaavya-VirtualBox:~/Desktop/25MCB1010$ ./mru.exe
Enter the number of pages: 12
Enter the page numbers: 2 5 1 0 1 0 0 1 0 1 5 2
Enter the number of frames: 4
2: 2 - - -
5: 2 5 - -
1: 2 5 1 -
0: 2 5 1 0
1: 2 5 1 0
0: 2 5 1 0
0: 2 5 1 0
1: 2 5 1 0
0: 2 5 1 0
5: 2 5 1 0
2: 2 5 1 0
Hits: 8
Faults: 4
```

**Optimal: ( Frames = 4 )**

Manual Implementation:

Optimal (4 frames)

Pages = 2 5 1 0 1 0 0 1 0 1 5 2

|   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 2 |   | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| 3 |   |   | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 4 |   |   | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|   | f | f | f | f | h | h | h | h | h | h | h |

Page faults = 4  
Page hits = 8  
Total = 12

Hit Ratio =  $8/12 = 66.67\%$   
Fault Ratio =  $4/12 = 33.33\%$

Code and Output:

```
bhaavya@bhaavya-VirtualBox: ~/Desktop/25MCB1010
#include <stdio.h>
#include <stdbool.h>
int main(){
    int n;
    printf("Enter the number of pages: ");
    scanf("%d",&n);
    int pages[n];
    printf("Enter the page numbers: ");
    for(int i=0;i<n;i++) scanf("%d",&pages[i]);
    int frame;
    printf("Enter the number of frames: ");
    scanf("%d",&frame);
    int frames[frame];
    for(int i=0;i<frame;i++) frames[i]=-1;
    int hit=0, fault=0;
    for(int i=0;i<n;i++){
        int page=pages[i];
        bool found=false;
        for(int j=0;j<frame;j++){
            if(frames[j] == page){
                hit++;
                found=true;
                break; } }
        if(!found){
            fault++;
            int replace=-1;
            for(int j=0;j<frame;j++){
                if(frames[j] == -1){replace = j; break; } }
            if(replace==-1){
                int next_use[frame];
                for(int j=0;j<frame;j++){
                    next_use[j]=n+1;
                    for(int k=i+1;k<n;k++){
                        if(frames[j] == pages[k]){next_use[j] = k; break;} } }
                int far_index=0;
                for(int j=1;j<frame;j++){
                    if(next_use[j]>next_use[far_index]) far_index=j; }
                replace=far_index; }
            frames[replace] = page; }
        printf("%d: ",page);
        for(int j=0;j<frame;j++){
            if(frames[j] == -1) printf("- ");
            else printf("%d ",frames[j]); }
        printf("\n"); }
    printf("Hits: %d\n",hit);
    printf("Faults: %d\n",fault);
-- INSERT --
```

```
bhaavya@bhaavya-VirtualBox:~/Desktop/25MCB1010$ vim optimal.c
bhaavya@bhaavya-VirtualBox:~/Desktop/25MCB1010$ gcc optimal.c -o optimal.exe
bhaavya@bhaavya-VirtualBox:~/Desktop/25MCB1010$ ./optimal.exe
Enter the number of pages: 12
Enter the page numbers: 2 5 1 0 1 0 0 1 0 1 5 2
Enter the number of frames: 4
2: 2 - - -
5: 2 5 - -
1: 2 5 1 -
0: 2 5 1 0
1: 2 5 1 0
0: 2 5 1 0
0: 2 5 1 0
1: 2 5 1 0
5: 2 5 1 0
2: 2 5 1 0
Hits: 8
Faults: 4
bhaavya@bhaavya-VirtualBox:~/Desktop/25MCB1010$
```